No Symmetric Width-16 Canonical Frame for the Lifted-Product Code $C^{\mathrm{LP}}_{16,9}$

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Abstract

Lee asks for the canonical width f of the $[[1088, 128, 16]]$ self-adjoint lifted-product code $C^{\mathrm{LP}}_{16,9}$, for which $f \in \{16, 18\}$ is known. The question reduces to whether the anticommutation graph between the 720 weight-16 X-type and 720 weight-16 Z-type logical classes has an induced matching of size 128. We determine the full automorphism group of this finite problem: it is $C_{16} \times C_2$, generated by the cyclic shift and the $X \leftrightarrow Z$ exchange, and it has exactly three involutions; this computation and all written proofs are checked in Lean 4. We show, by exhaustive computation validated with planted positive controls and a second, differently pruned search, that no width-16 canonical frame is fixed by any of them. Each search writes a proof certificate (its full search tree), and a checker written by a different AI model from a specification alone, which regenerates the problem data itself, accepts all three certificates; for the swap class the certificate check is also proved sound in Lean. Every width-16 frame, if one exists, therefore has trivial stabilizer, and the minimum width among frames with a nontrivial symmetry is 18. We also explain the block structure of the problem (the 3^{16} frames of one quarter, and why no frame splits along the coordinate blocks), show that the largest known partial frames (88 and 84 pairs) are far from any frame, and size the remaining unrestricted search. The source question itself remains open.

1 Result and motivating question

Source	Question	Answer here
J. Y. Lee, *Designing Group-Valued Codes with Full Regular Low-Weight Bases*, arXiv:2610.06820v1, 5 Oct. 2026, Prop. 9, §IV.4, App. C.3 [1]	Is the unrestricted canonical width $f(C^{\mathrm{LP}}_{16,9})$ equal to 16 or to 18?	Open. No width-16 frame has a nontrivial symmetry of the problem (Theorem 1); any width-16 frame is asymmetric.
Project follow-up (this paper)	Which symmetry classes of frames can achieve width 16?	None (Theorems 6 and 1).

Table 1: Question, source and answer.

Lee [1] defines the canonical width $f(C)$ of a quantum code as the smallest w for which C has a canonical (symplectic) basis of logical Pauli operators of weight at most w. The source places it in the hierarchy $d \le f_{\mathrm{span}} \le f \le f_{\mathrm{reg}} \le f_{\mathrm{rep}}$ and states $f_{\mathrm{span}} = 16$ and $f_{\mathrm{reg}} = 18$ for $C^{\mathrm{LP}}_{16,9}$ with its C_{16} action. Earlier work in this repository showed $f \in \{16, 18\}$ and reduced the question to a finite matching problem (§3).

Theorem 1 (main result)**.** *No width-*16 *canonical frame of* $C^{\mathrm{LP}}_{16,9}$ *is fixed by a nontrivial element of the automorphism group* $C_{16} \times C_2$ *of the frame problem. Every width-*16 *frame, if one exists, lies in a free orbit of* 32 *frames. In particular the minimum width over frames with a nontrivial symmetry is* 18*.*

The proof is computational: Theorem 6 identifies the group, Lemma 8 reduces the claim to three involution classes, and Table 2 gives the evidence for each. The T^8 case subsumes the source's $f_{\mathrm{reg}} = 18$ and the earlier exclusion of T^4-invariant frames [5], since every nontrivial subgroup of C_{16} contains T^8.

Evidence labels. [proof]: written proof. [certified]: computation that outputs an independently checkable certificate (a DRAT proof, or a search-tree certificate accepted by an independently written checker, §6). [computation]: exhaustive search by a program, cross-checked as stated, not certified. Every written proof has a Lean version (§10).

Contributions. New here: Theorem 6; Theorem 1 for the classes T^8, s and sT^8; the linear-algebra search of §6 with its validation; the structural results on blocks (Lemma 5 and §5: $Z = \pi(X)$, the 3^{16} count, no block splitting); the near-miss analysis of §7; the negative results of §8; the Lean formalization of §10, including Theorem 6; an independent implementation by a second agent, and proof certificates for the three classes accepted by an independently written checker (§6). Cited from earlier repository work [3, 4, 5]: the pools and the reduction, the T^4 exclusion, the partial frames and the relaxation experiments.

2 Background

The code. $C^{\mathrm{LP}}_{16,9}$ is the self-adjoint lifted product [2] of the 3×5 matrix over $\mathbb{F}_4[C_{16}]$ with coefficient array $((1,1,1,1,1),(1,1,\omega,0,\omega^2),(1,\omega^2,1,\omega,0))$ and exponent array $((0,0,0,0,0),(0,6,12,9,3),(0,12,6,2,14))$ [1, eq. (50)], expanded to binary by the trace representation $\rho(\omega) = \left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)$, $\rho(\omega^2) = \left(\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right)$. It has 480 independent checks of weight at most 9 in each sector, $n = 1088$ and $k = 128$; both CSS distances are 16 [4]. The checks are rebuilt independently by `verify_partial_frame.py`.

Definition 2. A *canonical frame* of a CSS code is a family $(\bar{X}_i, \bar{Z}_i)_{i=1}^{k}$ of logical operators in which $\bar{X}_i$ and $\bar{Z}_j$ anticommute iff $i = j$ and all other pairs commute. Its width is its largest operator weight, and $f(C)$ is the minimum width.

Coordinates and the cyclic action. Fix the width-18 frame as a basis of the logical space. Logical classes then have coordinates in $\mathbb{F}_2^{128}$, arranged as eight blocks (seeds) of 16 bits, and the shift T rotates each block by one position. With $R = \mathbb{F}_2[t]/(t^{16}+1) = \mathbb{F}_2[t]/((t+1)^{16})$, a local ring, the logical space is the free module R^8.

3 Reduction to an induced-matching problem (earlier work)

Proposition 3 ([computation]; [4])**.** *The nontrivial pure logical classes of weight at most* 17 *are exactly* 720 *X-classes and* 720 *Z-classes, all of weight* 16*, forming* 45 *free* T*-orbits per sector. Any canonical frame of width at most* 17 *yields a pure-sector frame drawn from these pools.*

The pool coordinates are `pools/xcoords.txt` and `pools/zcoords.txt`; index $16a + t$ is T^t applied to the a-th orbit representative. Write $G_{ij} = \langle x_i, z_j \rangle \in \mathbb{F}_2$.

Proposition 4 ([proof]). *$f(C^{\mathrm{LP}}_{16,9}) = 16$ iff there are $S, T \subseteq \{1, \dots, 720\}$ with $|S| = |T| = 128$ and $G[S,T]$ a permutation matrix, i.e. iff the bipartite anticommutation graph has an induced matching of size 128. No induced matching is larger, since its size is at most $\operatorname{rank}_{\mathbb{F}_2} G \le 128$.*

Proof. If $G[S,T]$ is a permutation matrix then $X_S Z_T^{\mathsf{T}}$ is invertible and the matched pairs form a canonical frame; the converse is Proposition 3. An induced matching of size m is an $m \times m$ identity submatrix of G. □

Lemma 5 ([computation]). *G is symmetric and T-invariant, and every pool orbit spans a free rank-one R-module. Moreover $z_i = \pi(x_i)$ for every index i, where π exchanges blocks $2 \leftrightarrow 4$ and $3 \leftrightarrow 5$; so $G_{ij} = x_i^{\mathsf{T}} \pi x_j$ and a frame is a pair of bases from one pool, dual for this symmetric form. (Symmetry is asserted by `satcheck/sat_symmetric.py` and checked in Lean; the module statement is computed in §8; $z_i = \pi(x_i)$ is checked by `theory/check_pi.py`.)*

Earlier results compared against below: frames containing at least 13 seed-4 X monomials do not exist, by a scan of all 3^{16} partner assignments [3]; T^4-invariant frames do not exist (DRAT-certified) [5]; explicit partial frames with 84 and 88 pairs exist (`partial-frame-84.json`, `partial-frame-88.json`); the basic degree-two moment relaxation is feasible at 128 [5]; and an aggregate residue-descent argument fails in enlarged pools [6].

4 The symmetry group

Theorem 6 ([computation], checked in Lean). *Let $\mathcal{G}$ be the group of pairs of bijections (π, π') of the index set with $G_{\pi i, \pi' j} = G_{ij}$, together with their compositions with the side exchange $s : (i,j) \mapsto (j,i)$, which preserves frames because G is symmetric. Then $\mathcal{G} = \langle T \rangle \times \langle s \rangle \cong C_{16} \times C_2$. Every linear map of the logical space preserving both pools induces an element of $\mathcal{G}$.*

Proof. Colour refinement [7] of the anticommutation graph (1440 vertices) stabilizes at the 90 T-orbits, or at 45 classes of size 32 when the two sides may be exchanged. After one vertex v is individualized, refinement reaches the discrete partition. Refinement is equivariant, so an automorphism fixing v fixes every vertex; hence $|\mathcal{G}|$ is at most the size of the colour class of v, which is 32. Both T and s are automorphisms. The computation is `explore/gram_wl.py`; coordinate permutations alone are bounded in the same way by `explore/auto_wl.py`. □

Corollary 7. *$\mathcal{G}$ has exactly three involutions, T^8, s and sT^8, and every nontrivial subgroup of $\mathcal{G}$ contains one of them.*

Lemma 8 ([proof]). *A frame is determined by its vertex sets (S,T). Writing τ for T^8 on indices: (i) the frame is T^8-fixed iff $\tau S = S$ and $\tau T = T$; (ii) it is s-fixed iff $S = T$; (iii) it is sT^8-fixed iff $T = \tau S$.*

Proof. The image of a frame under an element of $\mathcal{G}$ is the frame on the image vertex sets, and the induced matching on given vertex sets is unique. □

5 Evidence for Theorem 1

Linear-algebra search. Node counts of sharded runs include the shared top of the tree once per shard: the sT^8 search on 2 shards (50,401,313) and the certificate on 4 shards (50,404,337) both have 50,399,801 distinct nodes. The T^8 certificate is written by `la/la_gen.c` from the item instance, a different program and instance from the `la/la_dfs.c` run, so its count differs slightly.

Class	Linear-algebra search (§6)	Independent checks
T^8	none: 9,995,481 nodes; 810,784 with look-ahead. Planted controls found (seeds 1, 2).	Certificate: 9,996,689 nodes, accepted by the independent checker (§6); supplementary: Kissat cube-and-conquer on the frozen encoding, at the time of writing 243 pieces closed and 20 open (seven hard pieces split after 55 h each), none satisfiable
s	none: 69,180 nodes, 34 s. Planted control found.	Certificate: 69,180 nodes, accepted by the independent checker, and proved in Lean (§10); supplementary: vertex SAT encoding, planted control SAT, real instance undecided after 2 h (also with symmetry breaking)
sT^8	none: 50,401,313 nodes, 2 shards, ≈ 1.85 h each. Planted control: unconstrained search exceeded its time limit (40M nodes), but with any 1 to 48 of the 64 planted items fixed the planted frame is found in at most 82 nodes and verified.	Certificate: 50,404,337 nodes in 4 shards, accepted by the independent checker
T^4 (earlier)	none: 15,088 nodes	DRAT-certified [5]

Table 2: Evidence per involution class. Programs and instance generators: `la/`, `satcheck/`; run records in `RESTRICTED-FRAMES-20261031.md`, Follow-ups E and F.

The T^8 instance is written by `la/gen_instance.py` (SHA-256 prefix `60acef5f7b8a0bf0`) and searched by `la/la_dfs.c`; the s and sT^8 instances are written by `la/gen_items.py` and searched by `la/la_gen.c`. Frames found by any run are checked by `la/verify_frame.py` or `gen_items.py verify`.

Independent SAT check. `satcheck/sat_symmetric.py` shares no code with the search. It encodes Lemma 8 directly: choose 128 X-indices; each chosen X vector has exactly one chosen Z neighbour in G and vice versa, with T fixed by the class. Kissat [9] solves it, and `satcheck/check_model.py` decodes and checks any model.

Planted controls. A random T-invariant basis and its dual basis are appended to the pools as $8 + 8$ extra orbits. The planted frame lies in every class, and each method must find it.

Block-diagonal frames. [computation] No frame of the 64-dimensional subspace on seeds $\{0, 1, 4, 5\}$ or on $\{2, 3, 6, 7\}$ uses only pool vectors supported there (instances from `la/gen_instance.py support`, searched by `la/la_dfs.c`). [certified] No frame splits along the four seed-quarters (`quarters.py`, `support_cnf.py`, DRAT proofs checked by drat-trim; `results/quarters.json`). The quarter $\{0, 1\}$ alone carries exactly $3^{16} = 43{,}046{,}721$ frames of its subspace (full enumeration by the same program). [proof] This count has a short explanation: the pool vectors supported on blocks $\{0, 1\}$ are $e_{0,s}$, $e_{1,s}$ and $e_{0,s} + e_{1,s}$ on both sides, so the subspace splits orthogonally into 16 copies of $\mathbb{F}_2^2$, one for each position s, each with exactly 3 frames; on the other quarters a shift by 3 between the two pools leaves no frame. [proof; computation] More generally, no frame splits along any partition of the 8 blocks: if a frame contains a full frame of the coordinates of a block set B, the rest is a frame of the other blocks, and a complete search

finds a local frame only for $B \subseteq \{0,1,3,5,7\}$, whose complement contains blocks 2, 4 and 6 and so has none. Details and scripts: `theory/`.

6 Method: exact search on the residual identity

A family of 128 pairs (x_i, z_i) is a frame iff $\sum_i x_i z_i^{\mathsf{T}} = I$ over $\mathbb{F}_2$. The search state is a partial frame P together with the residual $R = I - \sum_P x_i z_i^{\mathsf{T}}$, the projection onto $C = \{z_i : i \in P\}^{\perp}$ along $\operatorname{span}\{x_i : i \in P\}$.

Lemma 9 (branching; [proof]). *If $R_{ck} = 1$, every completion contains a remaining pair with $x_c = z_k = 1$. Branching over those pairs, with earlier siblings excluded from later subtrees, visits each completion exactly once.*

Lemma 10 (rank pruning; [proof]). *Let X' (Z') be the unused pool vectors orthogonal to all chosen Z (X) vectors that belong to a still-admissible pair. A completion exists only if* $\operatorname{span} X' = C$ *and* $\operatorname{span} Z' = D$, *both of dimension* $128 - |P|$.

Lemma 11 (look-ahead; [proof]). *For an item with X-group a and Z-group b of size g, the child passes the rank test iff* $\operatorname{rank}(X' \cap b^{\perp}) \geq need - g$ *and* $\operatorname{rank}(Z' \cap a^{\perp}) \geq need - g$. *The pairing between these spans is then perfect automatically, so the look-ahead costs one span computation per group rather than per item.*

Validation. The general program `la/la_gen.c` reproduces the node counts of `la/la_dfs.c` exactly on the T^4 instance (15,088, with identical per-depth counts). Planted frames are found in every class tested. The quarter $\{0,1\}$ enumeration returns exactly 3^{16} frames.

Independent implementation. A separate agent (Codex, reported as based on GPT-6; exact version unavailable) wrote a second checker from a self-contained specification only (`SPEC.md` (pinned)), without access to this paper's code. It uses different methods: orbit reduction with coloured clique search, a weighted clique search for the swap classes, and a component-wise matching polynomial for counting. It passes all four known-answer tests, which we reran on our machine: planted frames found in T^8, s and sT^8; no T^4-invariant frame (complete, 25,327 nodes, 3 s); exactly 3^{16} frames on blocks $\{0,1\}$; none on blocks $\{2,3\}$. Every witness also passes our own checker. Its real T^8, s and sT^8 runs were all INCOMPLETE at their 2-hour limits (no frame found, search unfinished), so it does not confirm Theorem 1; it confirms the method-independent test answers and the T^4 result. Code and records: `independent-check/codex-gpt6/` (pinned).

Proof certificates. With option `-C`, `la/la_gen.c` writes the search tree: each node is a rank leaf, an empty leaf at a residual entry (c,k), or a branch at (c,k) listing its children. Specification: `CERT-SPEC.md`. A checker recomputes every node's state (chosen items, residual, live items) and confirms three things. The entry R_{ck} is 1. The children are exactly the live items covering (c,k), so with the residual identity and the minimal-index argument no completion is missed. Each leaf holds: an empty cover, or a live pairing matrix of rank below the remaining size. The certificates for T^8 (written from the item instance, with live-side pruning), s and sT^8 are in `certificates/`, with SHA-256 values in `certificates/MANIFEST.json`.

Independent certificate checker. ChatGPT (OpenAI; exact version not exposed by its runtime) wrote a checker in C++ from `CERT-SPEC.md` and `SPEC.md` only, without access to our search code, our reference checker or this paper. It regenerates the items from the raw pools and the class and requires them to equal the instance files. It passes the known-answer tests: it accepts the T^4 certificate, rejects three altered certificates and ten further mutations of its own, and finds and verifies the planted frame. It then accepts all three real certificates, including the cross-shard coverage checks. The node counts agree exactly with our runs: 69,180 (s, 3 s), 9,996,689 (T^8, 14 min) and 50,404,337 (sT^8, 67 min). We reran its checker on our machine with the same results. Code and records: `independent-check/chatgpt-certcheck/` (pinned). For the s class the certificate is also checked in Lean: a second checker by ChatGPT, written in Lean 4 and proved sound there, gives `no_swap_frame` (pinned): no 128-pair frame F of the real G satisfies $s(F) = F$. Its search soundness, item-table and frame-decomposition arguments are kernel-checked; two `native_decide` facts certify that the checker accepts `swap.cert.gz` and that the decoded vectors equal the pool data. We rebuilt it on our machine (`lake build`, 3,833 jobs, no errors, no `sorry`).

7 Near-miss partial frames are far from frames

Proposition 12 ([computation])**.** *Neither the* 88*-pair nor the* 84*-pair partial frame extends by a single pair. Some* 7 *of the* 88 *pairs already admit no completion. Every sampled subset of* 8 *or more pairs of the largest rank-compatible subsets (*40 *and* 41 *pairs) admits no completion.*

The driver is `la/nearmiss.py`, with fixed pairs passed to `la_gen -p`. As a control, planted frames are recovered after up to 60 of their pairs are removed. Large combinatorial partial frames are therefore far, in shared pairs, from rank-consistent ones; rank-consistent partial frames typically fail within 3 to 8 further choices.

8 The remaining problem

Knuth's estimator [8] puts the unrestricted search at roughly 2×10^{11}–2×10^{12} nodes, with a heavy tail. At a few thousand nodes per second this is 10^4–10^6 core-hours, comparable to large SAT-based proofs [10, 11].

The following were tested and do not reduce the search materially:

- parity checks $\#\{e \in F : z_e^{\mathsf{T}} L x_e = 1\} \equiv \operatorname{tr} L$ (the smallest odd-trace check found involves at least 1515 pairs; `explore/parity_checks.py`);
- small cocircuits of the pool matroids (unit functionals, which meet 66–88 pool vectors, are near-minimal; `explore/cocircuits.py`);
- the orbit-aggregated rank-inequality LP, whose optimum is exactly 128: `explore/rank_lp.py`;
- root normalization by a forced vector, with no net gain;
- orbits forced by rank, of which there are none;
- orderly generation modulo the 32-element group: a frame without symmetry lies in a free orbit of 32 frames, so this saves at most a factor of 32 and was not implemented;
- branching first on the blocks $2, 4, 6$ where local frames fail, or on the other blocks: Knuth estimates are 1–5 times the default fewest-candidates rule (`order-exp/RESULTS.md`).

Structured pieces a frame cannot contain. [proof; computation] If a frame contains d pairs spanning (U, U'), the remaining pairs form a frame of $(U'^{\perp}, U^{\perp})$. Using this: no frame contains a whole X-orbit paired one-to-one with a whole Z-orbit (167 of the 177 such orbit pairs fail by rank, the other 10 by complete search); 41 of the 45 orbits can never appear whole in a frame on either side, and orbits $9, 10, 11, 33$ remain undecided after time-limited searches. The 88-pair partial frame contains two of the excluded orbit pairs, which explains why it cannot be completed. None of this bounds how many vectors a frame takes from each orbit, so it is not an obstruction. Details: `theory/THEORY-NOTES-2.md`.

9 Open problems

Question 13 (source question [1])**.** Is $f(C^{\mathrm{LP}}_{16,9}) = 16$? Equivalently, does the anticommutation graph contain an asymmetric induced matching of size 128?

Follow-up directions raised by this work (not source-posed):

- What is the induced matching number of the anticommutation graph? It lies between 88 and 128.
- Why do rank-consistent partial frames fail after few choices while combinatorial ones reach 88? Can this gap be turned into an obstruction?
- Is there a small k with $|F \cap X_a| \le k$ for every frame F and orbit a? With $\sum_a |F \cap X_a| = 128$ over 45 orbits, a bound $k < 3$ would exclude frames.
- Is there a symmetrization principle for canonical frames of lifted-product codes over cyclic 2-group lifts? A naive residue descent fails [6]. The local ring R and its self-dual T^8-fixed submodule $(1 + t^8)R^8$ are natural tools.
- Let the *symmetric width* be the minimum width over frames with nontrivial stabilizer. It equals 18 here. When does it differ from f, for instance for $C^{\mathrm{LP}}_{20,10}$?

10 Lean formalization

The written proofs are checked in Lean 4 (toolchain `v4.36.0-rc1`, Mathlib `5f7b4198`) in `C16Frames.lean`. All declarations compile with no `sorry`, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The check is recorded in `verification.json`.

Paper statement	Lean declaration
Prop. 4, frame direction	`gram_one_iff_sum_outer` (pinned), `frame_of_perm_block` (pinned)
Prop. 4, size bound	`induced_matching_le_rank` (pinned), `induced_matching_le_dim` (pinned)
Lemma 8	`IsFrame` (pinned), `IsFrame.map` (pinned), `IsFrame.swap` (pinned), `fixed_swap_iff` (pinned), `fixed_half_iff` (pinned), `fixed_swapHalf_iff` (pinned)
Branching lemma (§6)	`exists_pair_of_residual_entry` (pinned), `first_member_unique` (pinned)
Rank-pruning lemma (§6)	`remaining_card_le_finrank` (pinned)
Look-ahead lemma (§6): dimension drop and perfect pairing	`perpOf` (pinned), `finrank_inf_perp_add` (pinned), `perfect_inf_perp` (pinned)
Bridge to the concrete pools (`Pools.lean`; data in `PoolData.lean`)	
Pools are T-orbits†	`X_rot` (pinned), `Z_rot` (pinned)
Lemma 5: G symmetric, T-invariant†	`G_symm` (pinned), `G_rot` (pinned)
Prop. 4 for these pools	`G_eq_mul` (pinned), `pool_matching_le_128` (pinned)
Symmetries act on pool frames†	`frame_rot` (pinned), `frame_swap` (pinned)
Lemma 8(iii) for these pools†	`tau8` (pinned), `fixed_sT8_iff` (pinned)
From the code to the pools (`Code.lean`, a port of `verify_partial_frame.py` and `audit_pools.py`)	
Code from eq. (50): 480+480 checks, weight ≤ 9, CSS†	`hx_length` (pinned), `hz_length` (pinned), `checks_weight_le_9` (pinned), `checks_commute` (pinned)
Width-18 frame is a canonical frame†	`canon_length` (pinned), `canon_logical` (pinned), `canon_gram` (pinned), `canon_width` (pinned)
Pools are distinct nontrivial weight-16 logicals with the recorded coordinates†	`pools_length` (pinned), `pools_weight_16` (pinned), `pools_logical` (pinned), `pools_distinct` (pinned), `coords_match` (pinned), `coords_nonzero` (pinned)
Physical anticommutation equals G†	`physical_pairing_eq_G` (pinned)
Mixed operators reduce to pure ones (`Mixed.lean`; Prop. 3, second sentence)	
Pairings, weights and types under the reduction	`ω_pure` (pinned), `card_supp_pure_le` (pinned), `kinds_differ_of_anticommute` (pinned), `pure_frame` (pinned)
The symmetry group (`Aut.lean`)	
Theorem 6: every automorphism is $T^k s^b$†	`aut_classification` (pinned), from `aut_of_certificate` (pinned) (refinement is equivariant, any finite graph) with `class_v0` (pinned), `dM_id` (pinned)
The 32 maps are distinct automorphisms; the graph is G†	`gmap_aut` (pinned), `gmap_distinct` (pinned), `adj_spec` (pinned)
Certificate check for the class s (ChatGPT, `CertGPT*.lean` (pinned))	
Theorem 1 for s: no s-fixed frame†	`no_swap_frame` (pinned); bridge to the (S,T) form, for any symmetric G: `no_swap_vertex_frame_of_no_pair` (pinned)

† Uses `native_decide` over the full 720×720 data, so it also trusts the Lean compiler (`Lean.ofReduceBool`). The pool data is generated from `pools/xcoords.txt` and `pools/zcoords.txt`, and the SHA-256 of both files is recorded in its header.

The abstract Lean statements: they concern any field, any pairing matrix and any finite index set. For the look-ahead lemma, Lean proves its two cores. The remaining step, that both spans are full exactly when the pairing has full rank, follows in one line because a pairing rank never

exceeds either dimension. The mixed reduction takes as a hypothesis that every operator of weight at most 17 has a stabilizer component, which follows from the computed $\mu_{XZ} = 26$ [4]. Theorem 6 is formalized for the anticommutation graph itself; its last sentence, that a linear map preserving both pools induces a graph automorphism, is immediate and not formalized. Not yet formalized: that computation and the completeness of the weight-16 enumeration [4] (every logical class of weight at most 17 lies in the pools); and the exhaustive computations, which remain [computation] or [certified] as labelled, except the s class, whose certificate is checked in Lean.

11 Reproducibility

Commands run from `restricted-frames-20261031/`; the C programs are compiled with `cc -O3 -march=native`. Search instances are regenerated by the listed command (the instances used for the certificates are committed in `certificates/`); the SHA-256 prefix (first 8 hex digits) identifies the file each recorded run used. Times are wall-clock on one core of a shared 2-core cloud machine unless stated.

Run	Command	Instance SHA-256	Result, time
T^8	`la/gen_instance.py order 2 inst/o2.txt`; `la/la_dfs -S 2 -s` i `inst/o2.txt`, $i = 0, 1$	`60acef5f`	no frame; 9,995,481 nodes; 801 s + 787 s
T^8, look-ahead	as above with `-l -f`	`60acef5f`	no frame; 810,784 nodes; 1,122 s + 1,011 s
s	`la/gen_items.py swap inst/g-swap.txt`; `la/la_gen -l inst/g-swap.txt`	`590699b0`	no frame; 69,180 nodes; 34 s
sT^8	`la/gen_items.py swapT8 inst/g-swapT8.txt`; `la/la_gen -l -S 2 -s` i `-D 1 inst/g-swapT8.txt`	`ab7d84bb`	no frame; 50,401,313 nodes; 6,679 s + 6,625 s
T^4 (validation)	`la/gen_items.py order 4 inst/g-o4.txt`; `la/la_gen inst/g-o4.txt`	`54d36d51`	no frame; 15,088 nodes; 2 s
Planted s	`la/gen_items.py --plant 1 swap inst/gp1-swap.txt`; `la/la_gen -l`	`0774c77f`	frame found, verified
Planted sT^8	`la/gen_items.py --plant 1 swapT8 inst/gp1-swapT8.txt`; `la/la_gen -l -p FIXED`	`757b7274`	frame found from 1–48 fixed items, verified
Swap SAT	`satcheck/sat_symmetric.py swap swap.cnf` [`--symbreak`]; `kissat swap.cnf`	CNF `588e61ed`; with symmetry breaking `05790b41`	UNKNOWN at 7,000 s
Certificates	`la/la_gen_c -l -C ../certificates/swap.cert.gz inst/g-swap.txt`; likewise `-S 2 -s` i `-D 1` on `inst/g-o2.txt` (T^8) and `-S 4 -s` i `-D 1` on `inst/g-swapT8.txt`	manifest	complete, no frame; 69,180 / 9,996,689 / 50,404,337 nodes; 27 s; 853 s ×2; ≈1,900 s ×4
Independent checker	`python3 run.py all` in `independent-check/chatgpt-certcheck/`	manifest	all three ACCEPT, tests pass; our rerun 3 s, 738 s, 3,675 s
Lean, s class	`cd lean && lake build CertGPTNoSwap`	manifest	`no_swap_frame`, no errors; 289 s

Run	Command	Instance SHA-256	Result, time
T^8 SAT (Mac)	`run-mac.sh`: Kissat cube-and-conquer, 4 parallel jobs on a Mac	frozen encoding	243 cubes UNSAT, 20 open
Lean	`cd lean && lake exe cache get && lake build`	in the header of `PoolData.lean`	0 errors, 3,833 jobs (with the s certificate check); `Pools` 675 s, `Code` 345 s

12 Availability

Repository links in this paper point to `master` of `https://github.com/alectionary/vibe-math`, where the work was merged with pull request 17.

Immutable review copies: the earlier work is pinned at commit `132f1161`, and the Lean files at commit `e534c654`. This paper's programs, instances and records were developed on branch `research/c16-structural-20261027T173245`, the head branch of PR 17.

References

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[4] *Independent low-weight pool enumeration.* `https://github.com/alectionary/vibe-math/blob/master/formalization/canonical-width/c16/INDEPENDENT-ENUMERATION.md`.

[5] *Structural comparison: exact restricted result, no quarter-turn invariant frame.* `https://github.com/alectionary/vibe-math/blob/master/formalization/canonical-width/c16/STRUCTURAL-COMPARISON.md`.

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